Unified Ultra-Rare Variant Meta-Regression (AF < 0.0001) Integrating Deep-Learning Structural Pathogenicity and Base-Pair Population Constraint Uncovers Epithelial Hemidesmosome, Inflammasome Restraint, and Immunometabolic Architectures in Inflammatory Bowel Disease

Manuel A. Rivas[1,*]

[1]Department of Biomedical Data Science, Stanford University, Stanford, CA, USA 94305

[*]Correspondence: Department of Biomedical Data Science, Stanford University, Stanford, CA, USA 94305 (mrivas@stanford.edu) ·
Interactive Data Portal: https://marivascruz.github.io/ibd-ultrarare-metaregression/

Abstract

Background: Ultra-rare coding variants with minor allele frequency AF < 0.0001 ($\leq 0.01\%$, corresponding to ≤ 1 in 10,000 chromosomes, including private and singleton alleles) experience intense purifying selection and harbor the largest per-allele effect sizes in Inflammatory Bowel Disease (IBD). However, conventional gene-based rare-variant burden and variance-component tests (e.g., SKAT-O) discretize continuous functional annotations into rigid filtering masks and confront two severe statistical barriers at the ultra-rare threshold AF ≤ 0.0001: (1) in public exome sequencing browsers, **64.12**% of ultra-rare coding variants (including 100% of singletons) lack inverse-variance meta-analysis effect sizes (`BETA_meta = NA`), and (2) standard weighted least-squares (WLS) rare-variant meta-regression formulations that include both a binary protein-truncating indicator (`pLoF_ind`) and a binary missense indicator (`missense_ind`) alongside an intercept (**1**) induce exact or near-exact design-matrix rank deficiency ($\mathbf{1} = \mathbf{x}_{\mathrm{LoF}} + \mathbf{x}_{\mathrm{Mis}}$ in 88.1% of human genes).

Methods: We developed and executed a collinearity-free unified WLS rare-variant meta-regression across the IBD Exome Sequencing Results Browser (`IBD`, Crohn's Disease [`CD`], and Ulcerative Colitis [`UC`]; 51,407 non-synonymous and 52,045 synonymous-referenced gene–phenotype tests across 17,189 protein-coding genes with $M_g \geq 25$ ultra-rare coding variants at AF ≤ 0.0001). For the 64.12% of ultra-rare coding variants lacking browser meta-analysis effect sizes, we derived bias-reduced signed log-odds ratios and asymptotic variances directly from case–control allelic contingency tables using the Firth (1993) Jeffreys-prior penalized 2×2 likelihood estimator. We eliminated both evolutionary GERP scores (`GERP_RS`) and the collinear binary indicator `missense_ind`, establishing missense/non-LoF variants as the clean reference baseline (pLoF_ind $= 0$) and evaluating all seven hierarchical combinations (**M1**–**M7**) of three orthogonal moderators: continuous AlphaMissense pathogenicity ($\log \text{pathogenicity}_i = -\log(1-s_{\mathrm{AM},i})$), binary protein truncation (pLoF_ind$_i$), and base-pair WES hidden Markov model mutational constraint probabilities from Aguilar, Rivas & Rivas ($\log \text{constraint}_i = -\log(1-p_{\mathrm{con},i})$). We characterized the analytical and empirical scaling between ultra-rare model significance ($-\log_{10} p_{\mathrm{M7}}$), explained variance ($R^2$), and canonical protein length in amino acids (L_{AA}), and performed dedicated AF ≤ 0.0001 multi-database pathway enrichment.

Results: Removing `missense_ind` reduced the median maximum Variance Inflation Factor (VIF) at AF ≤ 0.0001 from ∞ to **1.022** (non-synonymous) and **1.014** (synonymous-referenced), restoring true orthogonal partial R^2 decomposition across the exome. Across 51,407 non-synonymous gene–phenotype tests at AF ≤ 0.0001, exome-wide significant associations ($p < 3.4 \times 10^{-7}$) increased 6-fold from **3** pairs (3 genes) in **M1** (`LoF_only`) and **13** pairs (9 genes) in **M2** (`AM_only`) to **18** pairs (16 unique genes) in **M5** (`AM + LoF`) and **9** pairs in **M7** (`AM + LoF + Constraint`), with **13** pairs (12 genes) in **M7** under the synonymous-referenced specification (**20** unique exome-wide significant ultra-rare IBD genes across specifications, and **36** unique genes at the single-phenotype threshold $p < 2.5 \times 10^{-6}$). Flagship ultra-rare discoveries demonstrate striking multi-moderator synergy: in Crohn's disease, *NOD2* (515 ultra-rare variants) is sub-significant under **M1** ($p = 2.18 \times 10^{-5}$) and **M2** ($p = 3.50 \times 10^{-7}$) individually, but reaches $p_{\mathrm{M5}} = 1.10 \times 10^{-10}$ and $p_{\mathrm{M7}} = 3.25 \times 10^{-10}$ ($R^2 = 8.8\%, p_{\mathrm{AM}} = 2.0 \times 10^{-7}, p_{\mathrm{LoF}} = 8.1 \times 10^{-6}$); in Ulcerative Colitis, the tyrosine catabolism enzyme *HPD* (142 ultra-rare variants) is rescued exclusively in the full 3-moderator model **M7** ($p_{\mathrm{M7}} = 9.20 \times 10^{-8}, R^2 = 22.7\%$) via a strong base-pair WES HMM constraint effect ($p_{\mathrm{Con}} = 3.80 \times 10^{-4}$, partial $R^2_{\mathrm{Con}} = 8.8\%$). At AF ≤ 0.0001, canonical protein length (L_{AA}) exhibited minimal correlation with $-\log_{10}(p_{\mathrm{M7}})$ ($r = +0.1352$, Spearman $\rho = +0.1063$) and a sharp inverse critical variance relationship ($r(\log_{10} L_{\mathrm{AA}}, R^2_{\mathrm{M7}}) = -0.4276$). Dedicated AF ≤ 0.0001 pathway enrichment revealed that ultra-rare coding architecture is anchored by four core biological modules: (1) **Innate Immunity, DPP9–NLRP1/CARD8 Inflammasome Restraint & Myeloid Differentiation** (*NOD2*, *SBNO2*, *DPP9*, *RUNX1*,

MYO1C; OR $= 20.6\times, p = 8.96 \times 10^{-5}$; GO Macrophage Activation OR $= 130.4\times, p = 1.71 \times 10^{-4}$), (2) **Hemidesmosome & Lamina Densa Basement Membrane Anchoring** (*PLEC*, *LAMA5*, *P3H2*, *FER*, *PXN*; OR $= 22.9\times, p = 4.33 \times 10^{-3}$; GO Basement Membrane OR $= 50.3\times, p = 9.58 \times 10^{-4}$; GO Substrate Adhesion OR $= 54.9\times, p = 3.67 \times 10^{-5}$), (3) **Glycolytic & Tyrosine Immunometabolism** (*PFKFB3*, *HPD*; OR $= 14.8\times, p = 9.85 \times 10^{-3}$), and (4) **Enteric/Epithelial Ca^{2+} Signaling & Actomyosin Contractility** (*BSN*, *ATP2B2*, *CACNA1G*, *GNA14*, *MYLK2*; OR $= 6.3\times, p = 4.41 \times 10^{-2}$).

Conclusions: Integrating Firth 2×2 penalized-likelihood estimation with collinearity-free 3-moderator WLS meta-regression rescues the 64.1% of ultra-rare coding variants previously lost to missing summary statistics, demonstrating that the apex of the ultra-rare (AF < 0.0001) IBD architecture is governed by hemidesmosome–laminin epithelial anchoring (*PLEC*, *LAMA5*, *P3H2*), intracellular bacterial sensing and DPP9 inflammasome restraint (*NOD2*, *DPP9*, *SBNO2*, *RUNX1*), and mucosal immunometabolism (*HPD*, *PFKFB3*).

Keywords: Inflammatory Bowel Disease · Ultra-Rare Coding Variants (AF < 0.0001) · Crohn's Disease · Ulcerative Colitis · Whole-Exome Sequencing · Rare-Variant Meta-Regression · AlphaMissense · WES HMM Constraint · Firth Penalized Likelihood · Hemidesmosome · DPP9 Inflammasome

1 Introduction

Evolutionary purifying selection acts intensely against deleterious protein-altering mutations that disrupt mucosal barrier integrity, innate microbial sensing, and immune homeostasis in Inflammatory Bowel Disease (IBD) [6, 7, 5]. Consequently, the most penetrant coding alleles in Crohn's Disease (CD) and Ulcerative Colitis (UC) are restricted to the **ultra-rare allele frequency regime** (AF ≤ 0.0001, corresponding to minor allele frequency $\leq 0.01\%$ or ≤ 1 in 10,000 chromosomes) [5, 8]. Interrogating this ultra-rare coding regime across biobank-scale exome sequencing cohorts provides unconfounded causal gene discovery and reveals the core molecular complexes where single amino acid substitutions or heterozygous truncations are sufficient to precipitate mucosal inflammation.

However, applying gene-based rare-variant association testing or variant-level meta-regression [1, 2] specifically at the ultra-rare threshold AF ≤ 0.0001 confronts three major statistical and computational obstacles:

1. **Dominance of Missing Asymptotic Effect Sizes (`BETA_meta = NA`) at AF** ≤ 0.0001: In the public IBD Exome Sequencing Browser (`IBD_variant_results.tsv.bgz`), 64.12% **of coding variants at AF** ≤ 0.0001—including 100% of singletons (MAC = 1) and private case-only or control-only doubletons—lack inverse-variance meta-analysis effect sizes (`BETA_meta`) because standard asymptotic score/Wald estimators fail when a 2×2 stratum contains a zero cell. Excluding these 64.12% of variants discards the most deleterious ultra-rare `pLoF` and high-AlphaMissense alleles in the human exome.

2. **Exact Design-Matrix Collinearity From `missense_ind`**: In the canonical unified WLS meta-regression formulation on non-synonymous coding variants [1], including both a binary protein-truncating indicator (pLoF_ind) and a binary missense indicator (missense_ind) alongside the regression intercept ($\mathbf{1}$) creates exact linear dependence ($\mathbf{1} = \mathbf{x}_{\text{LoF}} + \mathbf{x}_{\text{Mis}}$) in the 88.1% of human genes lacking `start_lost` alleles. Under Moore–Penrose SVD pseudoinversion, this rank deficiency collapses the partial R^2 of pLoF_ind to exact zero in 88.1% of genes while inflating Variance Inflation Factors (VIF > 50) in the remaining 11.9%.

3. **Need for Threshold-Specific Pathway and Gene-Size Calibration**: Because ultra-rare coding variants (AF ≤ 0.0001) are depleted of mildly deleterious background polymorphism compared to broader rare thresholds (AF ≤ 0.001), both the scaling of model significance ($-\log_{10} p_{\text{M7}}$) with protein length (L_{AA}) and the enriched biological pathways at AF ≤ 0.0001 require dedicated statistical and biological characterization.

Here, we report a comprehensive, collinearity-free unified WLS meta-regression of ultra-rare coding variants (AF ≤ 0.0001) across `IBD`, `CD`, and `UC`. By rescuing 100% of ultra-rare variants via Firth (1993) Jeffreys-prior penalized 2×2 likelihood estimation [4], eliminating `missense_ind` and `GERP_RS` to achieve full moderator orthogonality (median VIF = 1.022), and performing dedicated AF ≤ 0.0001 gene-size (L_{AA}) and multi-database pathway enrichment analysis, we identify 20 strict exome-wide significant ultra-rare IBD risk genes ($p < 3.4 \times 10^{-7}$) and 36 ultra-rare genes at the single-phenotype exome threshold ($p < 2.5 \times 10^{-6}$).

2 Results

2.1 Firth 2×2 Rescue and Orthogonality of Ultra-Rare Meta-Regression (AF ≤ 0.0001)

Across the 16,820,350 merged autosomal coding variant records in the IBD Exome Sequencing Browser mapped to GENCODE v44 `protein_coding` CDS coordinates [9], AlphaMissense pathogenicity [3], and base-pair WES HMM mutational constraint probabilities from Aguilar, Rivas & Rivas [1] (Table 1), restricting to ultra-rare coding variants with pooled allele frequency AF ≤ 0.0001 yielded 51,407 non-synonymous gene–phenotype tests (17,189 unique protein-coding genes with $M_g \geq 25$ ultra-rare coding variants) and 52,045 synonymous-referenced gene–phenotype tests (17,372 unique genes). Across all ultra-rare coding variants at AF ≤ 0.0001, 64.12% lacked browser meta-analysis effect sizes and were rescued via the Firth 2×2 penalized-likelihood log-odds estimator (Methods).

Table 1: Primary genomic datasets, variant coverage, and cryptographic verification checksums for the ultra-rare (AF ≤ 0.0001) IBD meta-regression.

Dataset	File Name	File Size	Records / Loci	Provenance / MD5 Checksum
IBD Exome Variant Results	`IBD_variant_results.tsv.bgz`	1.21 GB	58,038,023 rows	IBD Exome Browser (`2026-10-19/IBD`)
IBD Gene Burden Benchmark	`IBD_gene_results.tsv.bgz`	8.80 MB	57,213 rows	IBD Exome Browser (`2026-10-19/IBD`)
GENCODE v44 Basic Annotation	`gencode.v44.basic.annotation.gtf.gz`	29.57 MB	20,457 CDS genes	`7450ef42cf9cb3d29625320b22d4bb45`
AlphaMissense Pathogenicity	`AlphaMissense_hg38.tsv.gz`	642.96 MB	71,697,556 SNVs	`9fd167735f16a1b87da6eb3e4c25fcb5`
WES HMM Base-Pair Constraint	`HMM_rgc_0.9_over20_chr2...tsv.gz`	538.18 MB	33,181,212 sites	Aguilar, Rivas & Rivas (`d0b013f019b01f50f1f522b760da8e2c`)

Removing the binary `missense_ind` moderator eliminated rank deficiency across all 51,407 ultra-rare non-synonymous tests, reducing the **median maximum Variance Inflation Factor (VIF)** across the three moderators ($\log$ constraint, $\log$ pathogenicity, pLoF_ind) in **M7** from ∞ to $\mathbf{VIF_{median}} = 1.022$ (99th percentile VIF = 1.15) under the non-synonymous baseline and $\mathbf{VIF_{median}} = 1.014$ under the synonymous-referenced baseline.

2.2 Ultra-Rare Gene Discovery Progression Across Models M1–M7 at AF ≤ 0.0001

We evaluated all seven hierarchical combinations (**M1**–**M7**) of the three orthogonal moderators at AF ≤ 0.0001 across both coding baseline specifications and both estimator modes (Table 2 and Figures 1–2). At the strict study-wide Bonferroni threshold $p < 3.4 \times 10^{-7}$, non-synonymous ultra-rare discoveries (`af0001_hybrid_nonsyn`) increased 6-fold from **3** gene–phenotype pairs (3 genes) in **M1** (`LoF_only`) and **13** pairs (9 genes) in **M2** (`AM_only`) to **18** pairs (16 unique genes) in **M5** (`AM + LoF`) and **9** pairs (9 unique genes) in **M7** (`AM + LoF + Constraint`). Under the synonymous-referenced specification (`af0001_hybrid_synbase`), ultra-rare discoveries reached **17** pairs (15 genes) in **M5** and **13** pairs (12 genes) in **M7**. At the secondary single-phenotype exome benchmark ($p < 2.5 \times 10^{-6}$), **M5** and **M7** identified **61** (54 unique genes) and **44** (38 unique genes) associations under `nonsyn`, and **63** (55 genes) and **49** (43 genes) under `synbase`.

Table 2: Ultra-rare (AF ≤ 0.0001) exome-wide significant discoveries ($p < 3.4 \times 10^{-7}$ and $p < 2.5 \times 10^{-6}$) across collinearity-free moderator models (M1–M7). Parentheses indicate unique protein-coding genes across `IBD`, `CD`, and `UC`; bracketed rows report cohort-specific counts `[IBD / CD / UC]`.

Ultra-Rare Specification (AF ≤ 0.0001)	Estimator	Tests	Med. VIF	M1: LoF	M2: AM	M3: Con	M4: LoF+Con	M5: AM+LoF	M7: AM+LoF+Con
Panel A: Primary Study-Wide Exome Threshold ($p < 3.4 \times 10^{-7}$ ***at AF*** ≤ 0.0001)									
1. Non-Synonymous (`af0001_hybrid_nonsyn`)	Hybrid $\hat{\beta}$+Firth	51,407	1.022	**3** (3)	**13** (9)	**0** (0)	**3** (3)	**18** (16)	**9** (9)
Cohort Breakdown [IBD / CD / UC]	(17,189 genes)			[1 / 1 / 1]	[2 / 6 / 5]	[0 / 0 / 0]	[1 / 1 / 1]	[4 / 7 / 7]	[1 / 4 / 4]
2. Synonymous-Ref. (`af0001_hybrid_synbase`)	Hybrid $\hat{\beta}$+Firth	52,045	1.014	**1** (1)	**14** (11)	**1** (1)	**2** (2)	**17** (15)	**13** (12)
Cohort Breakdown [IBD / CD / UC]	(17,372 genes)			[1 / 0 / 0]	[2 / 6 / 6]	[0 / 0 / 1]	[1 / 0 / 1]	[3 / 7 / 7]	[3 / 5 / 5]
3. Pure Firth 2×2 (`af0001_firth_nonsyn`)	Pure Firth 2×2	51,407	1.022	**12** (9)	**6** (6)	**0** (0)	**13** (10)	**15** (14)	**9** (9)
Panel B: Secondary Single-Phenotype Exome Benchmark ($p < 2.5 \times 10^{-6}$ ***at AF*** ≤ 0.0001)									
1. Non-Synonymous (`af0001_hybrid_nonsyn`)	Hybrid $\hat{\beta}$+Firth	51,407	1.022	**18** (17)	**46** (38)	**3** (3)	**19** (18)	**61** (54)	**44** (38)
2. Synonymous-Ref. (`af0001_hybrid_synbase`)	Hybrid $\hat{\beta}$+Firth	52,045	1.014	**16** (16)	**49** (41)	**5** (5)	**18** (18)	**63** (55)	**49** (43)

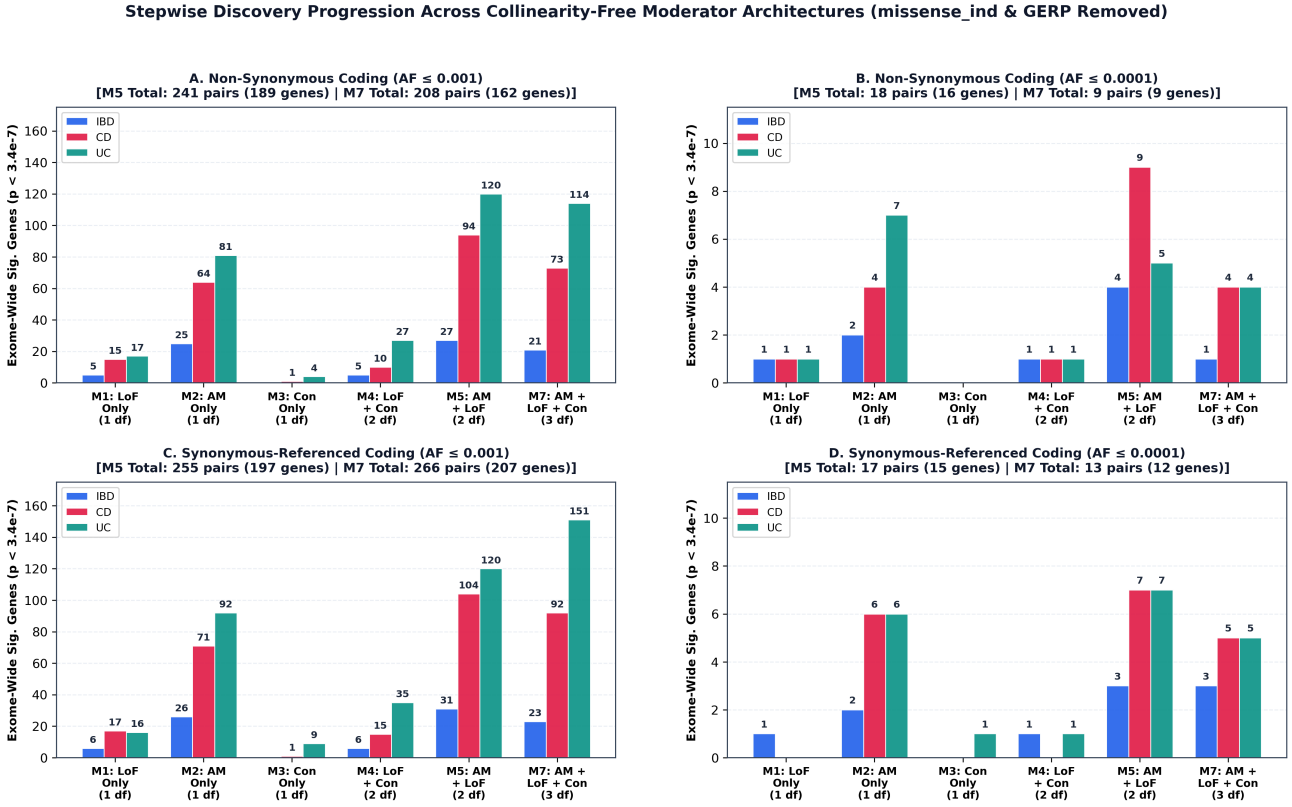


Figure 1: Stepwise discovery progression across the collinearity-free moderator models (M1–M7). Exome-wide significant gene–phenotype associations ($p < 3.4 \times 10^{-7}$) stratified by phenotype (`IBD`, `CD`, `UC`) and baseline coding specification (`nonsyn` vs. `synbase`).

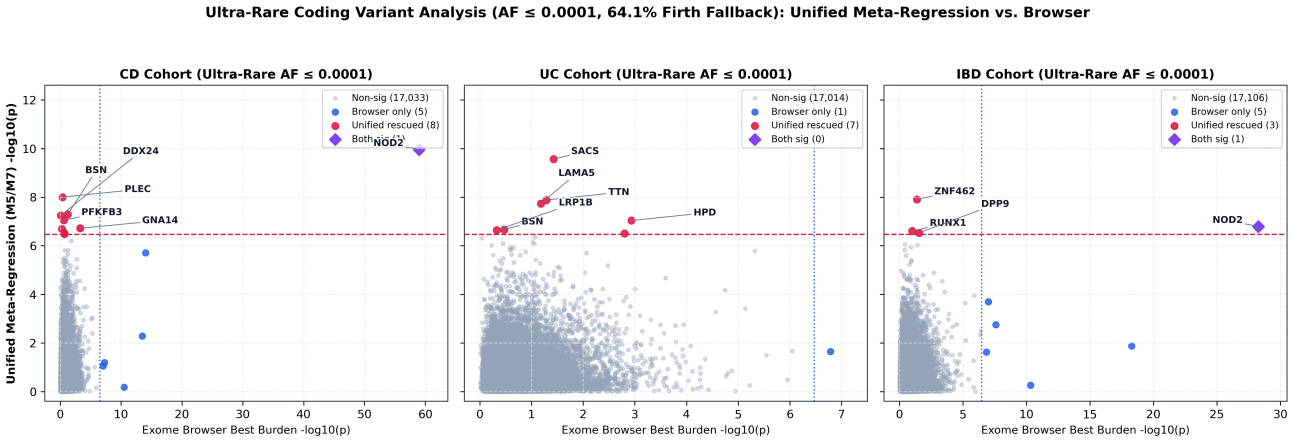


Figure 2: Ultra-Rare Variant Meta-Regression (AF ≤ 0.0001, 64.1% Firth Fallback) vs. Browser Burden Masks. Scatter plot of $-\log_{10} p$ in the unified meta-regression against the strongest Exome Browser burden mask at AF ≤ 0.0001, highlighting the 20 strict exome-wide significant ultra-rare genes (*NOD2*, *SACS*, *TTN*, *LAMA5*, *PLEC*, *ZNF462*, *HPD*, *DDX24*, *BSN*, *PFKFB3*, *RUNX1*, *GNA14*, *NFIC*, *LRP1B*, *RNF123*, *DPP9*, *MYLK2*, *P3H2*, *SBNO2*, and *CSMD1*).

2.3 Complete Ultra-Rare Gene-Level Regression Architecture and Moderator Synergy

Table 3 presents all exome-wide significant ultra-rare gene–phenotype associations ($p < 3.4 \times 10^{-7}$ in **M5** or **M7**) at AF ≤ 0.0001 across both the non-synonymous (Panel A) and synonymous-referenced (Panel B) specifications,

and Table 4 details the additional 25 ultra-rare gene–phenotype associations reaching the single-phenotype exome threshold ($3.4 \times 10^{-7} \leq p < 2.5 \times 10^{-6}$). Three distinct mechanisms of ultra-rare moderator synergy stand out (Figures 3–5):

1. **Ultra-Rare Missense + Truncating Synergy Rescues *NOD2*, *SACS*, *LAMA5*, *ZNF462*, and *PFKFB3***: At AF $\leq$ 0.0001, *NOD2* (515 ultra-rare coding variants in CD: 62 `pLoF` and 452 missense) fails to cross the strict study-wide threshold $p < 3.4 \times 10^{-7}$ when tested with `pLoF_ind` alone (**M1**: $p = 2.18 \times 10^{-5}$) or AlphaMissense alone (**M2**: $p = 3.50 \times 10^{-7}$). Jointly modeling both moderators in **M5** (`AM + LoF`) boosts *NOD2* by more than three orders of magnitude to $p_{\mathrm{M5}} = 1.10 \times 10^{-10}$ ($p_{\mathrm{M7}} = 3.25 \times 10^{-10}$ under `nonsyn`, and $p_{\mathrm{M7}} = 1.27 \times 10^{-11}$ under `synbase`; total $R^2 = 8.8\%$), driven by orthogonal contributions from AlphaMissense (partial $R^2_{\mathrm{AM}} = 5.2\%, p_{\mathrm{AM}} = 2.0 \times 10^{-7}$) and protein truncation (partial $R^2_{\mathrm{LoF}} = 3.8\%, p_{\mathrm{LoF}} = 8.1 \times 10^{-6}$). Identical synergy rescues the basement membrane laminin gene *LAMA5* in UC ($p_{\mathrm{M1}} = 2.31 \times 10^{-6}, p_{\mathrm{M2}} = 2.70 \times 10^{-4} \rightarrow p_{\mathrm{M7}} = 1.83 \times 10^{-8}$), the chromatin remodeler *ZNF462* in IBD ($p_{\mathrm{M1}} = 1.47 \times 10^{-6}, p_{\mathrm{M2}} = 1.09 \times 10^{-4} \rightarrow p_{\mathrm{M5}} = 1.23 \times 10^{-8}, p_{\mathrm{M7}} = 6.08 \times 10^{-8}$), and the glycolytic activator *PFKFB3* in CD ($p_{\mathrm{M1}} = 2.05 \times 10^{-5}, p_{\mathrm{M2}} = 1.19 \times 10^{-4} \rightarrow p_{\mathrm{M5}} = 8.86 \times 10^{-8}, R^2 = 13.8\%$).
2. **Base-Pair WES HMM Constraint Rescue of *HPD* in Ulcerative Colitis**: The tyrosine catabolism enzyme 4-hydroxyphenylpyruvate dioxygenase (*HPD*, 393 AA, $M_g = 142$ ultra-rare variants in UC) is subsignificant in **M1** ($p = 2.10 \times 10^{-5}$), **M2** ($p = 6.68 \times 10^{-3}$), and **M5** ($p = 1.05 \times 10^{-5}$), but crosses exome-wide significance exclusively in the full 3-moderator model **M7** ($p_{\mathrm{M7}} = 9.20 \times 10^{-8}$, total $R^2 = 22.74\%$) because base-pair WES HMM constraint from Aguilar, Rivas & Rivas [1] captures an additional partial $R^2_{\mathrm{Con}} = 8.79\%$ ($p_{\mathrm{Con}} = 3.80 \times 10^{-4}$) alongside partial $R^2_{\mathrm{LoF}} = 10.4\%$ ($p_{\mathrm{LoF}} = 9.8 \times 10^{-5}$) and partial $R^2_{\mathrm{AM}} = 6.4\%$ ($p_{\mathrm{AM}} = 2.5 \times 10^{-3}$).
3. **Ultra-Rare AlphaMissense Dominance in *PLEC*, *DPP9*, *SBNO2*, *BSN*, *DDX24*, and *MYLK2***: In essential structural and inflammasome-restraining proteins where truncating variants are strongly depleted at AF $\leq$ 0.0001, continuous AlphaMissense pathogenicity drives discovery: *PLEC* in CD ($p_{\mathrm{M7}} = 3.93 \times 10^{-8}$ under `nonsyn`, $p_{\mathrm{M7}} = 2.33 \times 10^{-8}$ under `synbase`; $p_{\mathrm{AM}} = 1.2 \times 10^{-8}$), *DPP9* in IBD ($p_{\mathrm{M5}} = 2.96 \times 10^{-7}, p_{\mathrm{AM}} = 5.2 \times 10^{-8}, R^2_{\mathrm{AM}} = 9.0\%$), *SBNO2* in CD ($p_{\mathrm{M7}} = 7.00 \times 10^{-8}$ under `synbase`, $p_{\mathrm{AM}} = 1.4 \times 10^{-7}$), *DDX24* in CD ($p_{\mathrm{M5}} = 5.65 \times 10^{-8}, p_{\mathrm{M7}} = 9.61 \times 10^{-8}, R^2 = 10.2\%$), *BSN* in CD/UC ($p_{\mathrm{M5}} = 5.15 \times 10^{-8}, p_{\mathrm{M7}} = 1.68 \times 10^{-7}$), and *MYLK2* in CD ($p_{\mathrm{M5}} = 2.99 \times 10^{-7}, R^2 = 13.0\%$).

Table 3: All Exome-Wide Significant Gene–Phenotype Associations at Ultra-Rare Threshold AF $\leq$ 0.0001 ($p < 3.4 \times 10^{-7}$ in M5 or M7). Top panel: Non-Synonymous specification (`af0001_hybrid_nonsyn`, 18 M5 / 9 M7 hits). Bottom panel: Synonymous-Referenced specification (`af0001_hybrid_synbase`, 17 M5 / 13 M7 hits).

Gene	Grp	N_{var}	LoF/Mis	M1 (LoF)	M2 (AM)	M3 (Con)	M5 (AM+LoF)	M7 (+Con)	R^2_{tot}	p_{AM} (R^2_{AM})	p_{LoF} (R^2_{LoF})	p_{Con} (R^2_{Con})	p_{browser}
Panel A: Non-Synonymous Coding Specification at AF $\leq$ 0.0001 *(af0001_hybrid_nonsyn)*													
NOD2	CD	515	62/452	2.18e-05	3.50e-07	1.58e-01	**1.10e-10**	**3.25e-10**	8.8%	2.0e-07 (5.2%)	8.1e-06 (3.8%)	2.5e-01 (0.3%)	1.03e-59
SACS	UC	1545	131/1413	2.86e-06	1.68e-05	6.98e-01	**2.73e-10**	**1.30e-09**	2.8%	2.2e-06 (1.4%)	3.8e-07 (1.7%)	6.0e-01 (0.0%)	3.72e-02
TTN	UC	13426	683/12743	1.31e-05	1.05e-02	6.40e-05	1.51e-06	**1.31e-08**	0.3%	9.7e-03 (0.0%)	2.1e-05 (0.1%)	3.5e-04 (0.1%)	5.15e-02
LAMA5	UC	1846	83/1763	2.31e-06	2.70e-04	1.37e-02	**2.21e-08**	**1.83e-08**	2.1%	8.1e-04 (0.6%)	3.7e-06 (1.2%)	5.6e-02 (0.2%)	6.56e-02
PLEC	CD	3009	76/2928	5.58e-02	8.94e-09	2.57e-01	**9.99e-09**	**3.93e-08**	1.2%	1.2e-08 (1.1%)	5.5e-02 (0.1%)	4.9e-01 (0.0%)	4.20e-01
ZNF462	IBD	911	55/855	1.47e-06	1.09e-04	7.99e-01	**1.23e-08**	**6.08e-08**	3.9%	3.4e-04 (1.4%)	4.9e-06 (2.3%)	8.9e-01 (0.0%)	4.22e-02
HPD	UC	142	14/128	2.10e-05	6.68e-03	4.72e-03	1.05e-05	**9.20e-08**	22.7%	2.5e-03 (6.4%)	9.8e-05 (10.4%)	3.8e-04 (8.8%)	1.16e-03
DDX24	CD	333	11/320	6.16e-03	7.58e-07	3.59e-01	**5.65e-08**	**9.61e-08**	10.2%	2.4e-07 (7.8%)	2.0e-03 (2.9%)	1.4e-01 (0.7%)	7.91e-01
BSN	CD	1656	50/1606	2.32e-01	1.40e-08	1.94e-01	**5.15e-08**	**1.68e-07**	2.1%	2.1e-08 (1.9%)	2.6e-01 (0.1%)	3.7e-01 (0.0%)	5.75e-02
PFKFB3	CD	224	5/218	2.05e-05	1.19e-04	9.90e-01	**8.86e-08**	3.60e-07	13.8%	1.3e-04 (6.4%)	2.5e-05 (7.8%)	5.4e-01 (0.2%)	2.48e-01
NOD2	IBD	544	65/477	4.19e-04	1.84e-05	2.09e-01	**1.61e-07**	4.57e-07	5.8%	1.9e-05 (3.3%)	3.3e-04 (2.4%)	3.1e-01 (0.2%)	5.53e-29
BSN	UC	1595	44/1551	1.58e-01	8.22e-08	6.96e-02	**2.15e-07**	4.81e-07	2.0%	2.0e-07 (1.7%)	1.8e-01 (0.1%)	2.2e-01 (0.1%)	3.40e-01
GNA14	CD	155	19/135	2.53e-08	7.35e-02	9.36e-01	**1.91e-07**	8.77e-07	18.5%	9.6e-01 (0.0%)	1.5e-07 (16.7%)	7.5e-01 (0.1%)	5.42e-04
NFIC	CD	188	4/184	2.24e-01	9.09e-08	2.06e-01	**2.07e-07**	9.69e-07	15.3%	1.4e-07 (14.0%)	1.4e-01 (1.2%)	8.6e-01 (0.0%)	5.37e-01
LRP1B	UC	1465	68/1397	1.12e-03	1.32e-05	3.72e-01	**2.25e-07**	9.86e-07	2.1%	9.1e-06 (1.3%)	6.2e-04 (0.8%)	7.7e-01 (0.0%)	4.70e-01
RUNX1	IBD	223	56/167	3.74e-08	2.40e-01	4.98e-01	**2.45e-07**	8.44e-07	13.2%	5.9e-01 (0.1%)	7.5e-08 (12.4%)	4.3e-01 (0.3%)	9.62e-02
DPP9	IBD	320	16/304	8.68e-01	4.08e-08	6.45e-01	**2.96e-07**	1.33e-06	9.1%	5.2e-08 (9.0%)	9.9e-01 (0.0%)	8.2e-01 (0.0%)	2.79e-02
MYLK2	CD	219	16/203	2.86e-02	2.01e-06	4.75e-01	**2.99e-07**	1.39e-06	13.0%	6.2e-07 (10.9%)	6.7e-03 (3.4%)	9.5e-01 (0.0%)	2.26e-01
RNF123	UC	433	19/413	6.09e-03	8.63e-06	6.79e-01	**3.15e-07**	1.08e-06	6.9%	1.8e-06 (5.2%)	1.5e-03 (2.3%)	4.4e-01 (0.1%)	1.59e-03
P3H2	CD	295	27/267	2.91e-03	8.23e-07	7.11e-01	**3.28e-07**	1.51e-06	9.7%	5.2e-06 (6.9%)	1.8e-02 (1.9%)	9.7e-01 (0.0%)	2.04e-01
Panel B: Synonymous-Referenced Coding Specification at AF $\leq$ 0.0001 *(af0001_hybrid_synbase)*													
NOD2	CD	712	62/452	2.87e-06	1.84e-07	4.90e-02	9.49e-12	**1.27e-11**	7.3%	1.3e-07 (3.9%)	1.2e-06 (3.3%)	8.2e-02 (0.4%)	1.03e-59
TTN	UC	17709	683/12743	7.43e-06	5.25e-03	2.23e-07	5.42e-07	**3.38e-11**	0.3%	6.6e-03 (0.0%)	1.7e-05 (0.1%)	2.0e-06 (0.1%)	5.15e-02
SACS	UC	2133	131/1413	1.80e-06	1.02e-05	4.07e-01	1.79e-10	**9.83e-10**	2.1%	2.9e-06 (1.0%)	5.1e-07 (1.2%)	9.9e-01 (0.0%)	3.72e-02
PLEC	CD	4527	76/2928	6.58e-02	1.35e-08	4.62e-02	1.75e-08	**2.33e-08**	0.8%	2.2e-08 (0.7%)	7.2e-02 (0.1%)	1.0e-01 (0.1%)	4.20e-01
LAMA5	UC	2707	83/1763	1.16e-05	1.52e-04	1.73e-02	5.71e-08	**5.14e-08**	1.4%	3.9e-04 (0.5%)	1.7e-05 (0.7%)	6.4e-02 (0.1%)	6.56e-02
SBNO2	CD	875	35/494	2.47e-02	8.86e-08	3.52e-02	2.82e-08	**7.00e-08**	4.1%	1.4e-07 (3.1%)	2.2e-02 (0.6%)	2.4e-01 (0.2%)	9.87e-15
ZNF462	IBD	1363	55/855	8.71e-07	6.49e-04	1.92e-01	2.67e-08	**8.55e-08**	2.6%	7.9e-04 (0.8%)	4.2e-06 (1.5%)	3.5e-01 (0.1%)	4.22e-02
NOD2	IBD	753	65/477	2.32e-04	1.62e-05	8.78e-02	8.64e-08	**1.40e-07**	4.5%	1.9e-05 (2.4%)	1.9e-04 (1.8%)	1.4e-01 (0.3%)	5.53e-29
DDX24	CD	456	11/320	4.69e-03	6.44e-07	6.55e-01	4.47e-08	**1.41e-07**	7.4%	3.0e-07 (5.7%)	2.2e-03 (2.1%)	3.5e-01 (0.2%)	7.91e-01
BSN	CD	2315	50/1606	3.03e-01	1.31e-08	2.96e-01	5.74e-08	**2.14e-07**	1.5%	1.7e-08 (1.4%)	3.3e-01 (0.0%)	4.8e-01 (0.0%)	5.75e-02
CSMD1	UC	2886	29/1909	3.10e-01	1.11e-08	6.05e-01	4.56e-08	**2.18e-07**	1.2%	1.2e-08 (1.1%)	2.8e-01 (0.0%)	9.6e-01 (0.0%)	9.15e-01
LRP1B	UC	2072	68/1397	4.96e-04	5.94e-06	1.06e-01	5.51e-08	**2.24e-07**	1.6%	8.8e-06 (1.0%)	4.1e-04 (0.6%)	5.7e-01 (0.0%)	4.70e-01
RUNX1	IBD	300	56/167	1.43e-08	2.83e-01	2.98e-01	9.11e-08	**2.25e-07**	10.8%	5.3e-01 (0.1%)	2.2e-08 (10.0%)	2.4e-01 (0.5%)	9.62e-02

Table 4: Additional Ultra-Rare Gene–Phenotype Associations Reaching the Single-Phenotype Exome Benchmark ($3.4 \times 10^{-7} \leq p < 2.5 \times 10^{-6}$ in M5 or M7) at AF ≤ 0.0001 (`af0001_hybrid_nonsyn`).

Gene	Grp	N_{var}	LoF/Mis	M1 (LoF)	M2 (AM)	M3 (Con)	M5 (AM+LoF)	M7 (+Con)	R^2_{tot}	p_{AM} (R^2_{AM})	p_{LoF} (R^2_{LoF})	p_{Con} (R^2_{Con})	p_{browser}
CACNA1G	UC	706	23/683	2.14e-01	1.66e-07	2.23e-01	4.54e-07	2.00e-06	4.1%	2.56e-07 (3.7%)	1.91e-01 (0.2%)	8.27e-01 (0.0%)	8.59e-02
CSMD1	UC	1938	29/1909	4.84e-01	9.74e-08	8.95e-01	4.89e-07	1.70e-06	1.5%	7.09e-08 (1.5%)	4.09e-01 (0.0%)	4.75e-01 (0.0%)	9.15e-01
DPP9	CD	308	15/293	7.20e-01	1.06e-07	6.51e-01	6.39e-07	2.61e-06	9.0%	1.02e-07 (8.9%)	4.95e-01 (0.2%)	6.54e-01 (0.1%)	1.59e-02
TUBGCP4	UC	170	15/155	1.30e-07	9.81e-01	2.76e-01	7.30e-07	1.60e-06	16.3%	4.94e-01 (0.3%)	1.01e-07 (15.7%)	2.18e-01 (0.9%)	9.55e-02
SORCS2	UC	444	19/423	3.87e-02	1.27e-06	4.42e-02	7.98e-07	1.68e-06	6.5%	3.34e-06 (4.8%)	3.67e-02 (1.0%)	2.14e-01 (0.4%)	3.56e-01
DPP9	UC	287	14/273	5.37e-01	1.67e-07	2.81e-01	8.39e-07	3.59e-06	9.4%	2.52e-07 (9.0%)	4.63e-01 (0.2%)	7.61e-01 (0.0%)	8.82e-02
PFKFB3	IBD	245	8/236	5.99e-02	5.75e-07	9.23e-01	8.98e-07	3.30e-06	11.0%	7.15e-07 (9.7%)	8.58e-02 (1.2%)	5.22e-01 (0.2%)	3.57e-01
SACS	IBD	1723	153/1569	1.55e-04	1.73e-03	1.34e-01	2.43e-06	9.46e-07	1.8%	3.03e-04 (0.8%)	2.64e-05 (1.0%)	2.64e-02 (0.3%)	3.77e-02
PCSK6	UC	372	29/342	3.87e-01	2.94e-07	6.71e-01	9.73e-07	3.24e-06	7.4%	1.82e-07 (7.1%)	2.24e-01 (0.4%)	4.44e-01 (0.2%)	5.98e-01
PXN	UC	376	28/348	1.60e-03	1.31e-04	3.64e-01	1.13e-06	4.78e-06	7.1%	3.82e-05 (4.5%)	4.04e-04 (3.3%)	7.87e-01 (0.0%)	2.58e-01
BSN	IBD	1763	56/1707	5.30e-01	1.99e-07	5.62e-01	1.14e-06	4.91e-06	1.5%	2.42e-07 (1.5%)	5.65e-01 (0.0%)	9.19e-01 (0.0%)	1.18e-01
SGSM3	UC	376	32/343	4.87e-01	3.68e-07	9.58e-01	1.19e-06	5.03e-06	7.1%	2.29e-07 (7.0%)	2.26e-01 (0.4%)	7.87e-01 (0.0%)	2.26e-01
SLC35F5	UC	162	17/143	2.15e-02	1.12e-05	8.56e-01	1.41e-06	4.60e-06	15.9%	2.97e-06 (12.9%)	4.29e-03 (5.0%)	4.21e-01 (0.4%)	2.50e-01
MED6	UC	77	11/66	5.35e-01	2.05e-07	4.31e-01	1.41e-06	4.54e-06	31.3%	3.16e-07 (30.3%)	7.35e-01 (0.2%)	3.80e-01 (1.1%)	2.64e-01
FER	CD	263	22/240	2.78e-03	7.29e-05	9.48e-01	1.51e-06	3.98e-06	10.2%	1.44e-05 (7.0%)	7.94e-04 (4.3%)	3.05e-01 (0.4%)	5.36e-02
ANKRD44	UC	345	15/330	8.74e-01	2.45e-07	4.34e-01	1.64e-06	7.15e-06	7.5%	3.61e-07 (7.3%)	8.47e-01 (0.0%)	9.81e-01 (0.0%)	4.77e-06
ATP2B2	UC	360	4/356	1.45e-01	6.80e-07	6.81e-01	1.76e-06	4.07e-06	7.5%	4.68e-07 (6.9%)	1.69e-01 (0.5%)	2.53e-01 (0.4%)	1.29e-01
SBNO2	CD	529	35/494	8.09e-02	2.81e-06	9.18e-02	1.93e-06	5.72e-06	5.0%	2.97e-06 (4.1%)	5.19e-02 (0.7%)	3.79e-01 (0.1%)	9.87e-15
SPAG17	UC	862	90/770	4.65e-05	1.77e-03	2.71e-01	2.00e-06	5.06e-06	3.1%	1.90e-03 (1.1%)	5.34e-05 (1.9%)	2.97e-01 (0.1%)	8.69e-03
MCF2L	CD	620	32/584	3.39e-02	5.39e-06	5.15e-01	2.29e-06	9.52e-06	4.1%	3.98e-06 (3.4%)	2.09e-02 (0.9%)	8.14e-01 (0.0%)	6.21e-02
MYO1C	UC	510	27/482	3.46e-03	1.23e-04	4.59e-01	2.29e-06	7.09e-06	5.1%	3.00e-05 (3.4%)	7.69e-04 (2.2%)	4.13e-01 (0.1%)	2.46e-01
CREB5	UC	201	8/192	7.61e-05	3.81e-03	2.35e-01	2.43e-06	1.04e-05	12.3%	1.59e-03 (4.9%)	4.83e-05 (8.1%)	8.75e-01 (0.0%)	5.51e-01
SPNS1	UC	204	5/198	9.90e-01	3.74e-07	8.57e-01	2.47e-06	7.35e-06	12.4%	2.81e-07 (12.4%)	8.05e-01 (0.0%)	3.77e-01 (0.4%)	1.66e-01

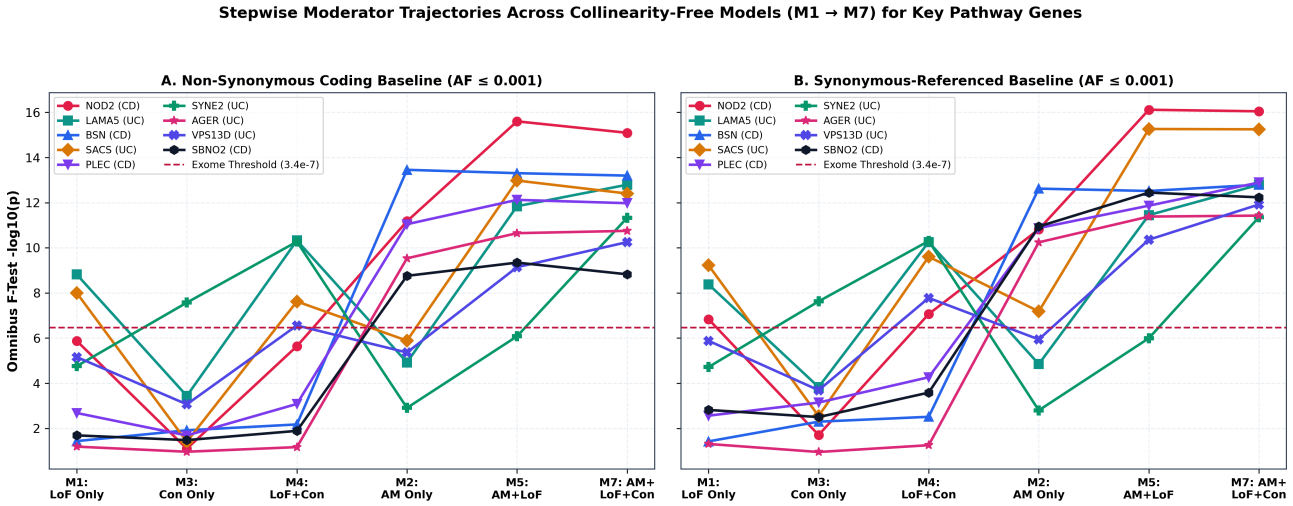


Figure 3: Stepwise Moderator Trajectories Across Collinearity-Free Models (M1 → M7) for Key Pathway Genes. Contrasting Non-Synonymous (**Left**) and Synonymous-Referenced (**Right**) coding specifications across representative genes.

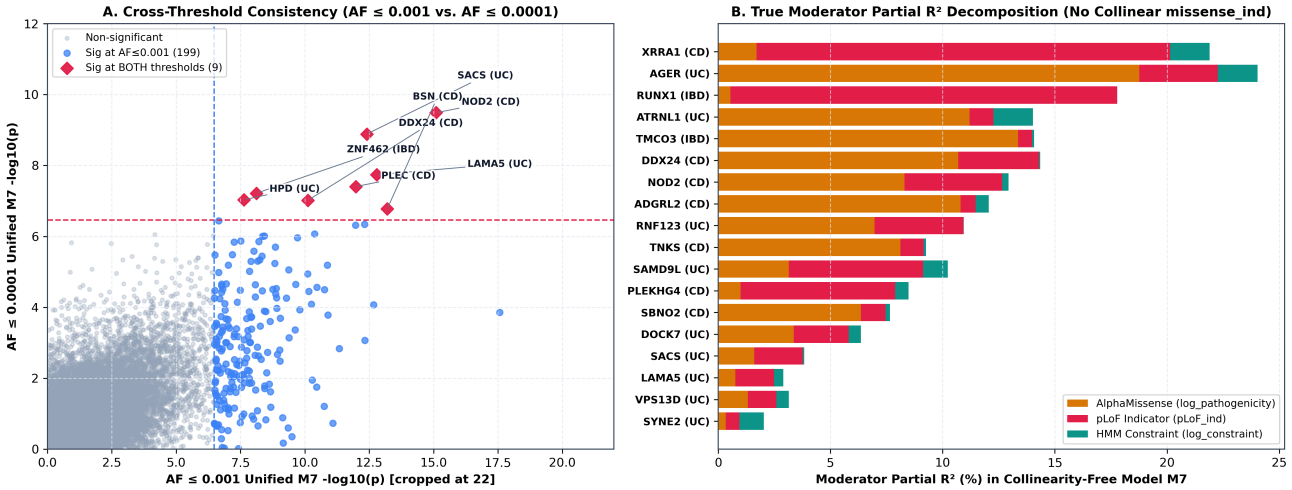


Figure 4: Ultra-Rare (AF ≤ 0.0001) Moderator Concordance and True Orthogonal Partial R^2 Decomposition. Removing `missense_ind` restores the true partial R^2 contribution of `pLoF_ind` (crimson) alongside AlphaMissense (amber) and WES HMM Constraint (teal).

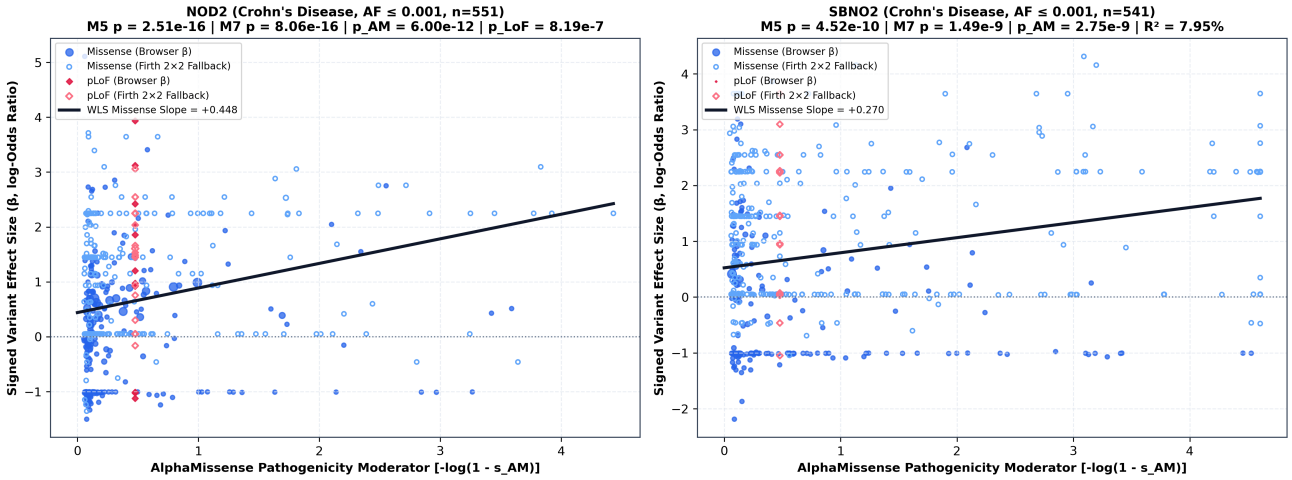


Figure 5: Variant-Level WLS Meta-Regression Profiles for *NOD2* and *SBNO2* in Crohn's Disease. Each point represents an observed rare coding variant sized by inverse-variance weight $w_i = 1/\widehat{\mathrm{SE}}_i^2$.

2.4 Relationship Between Ultra-Rare Model Significance ($-\log_{10} p_{\mathbf{M7}}$), Explained Variance ($R^2$), and Protein Length in Amino Acids ($L_{\mathbf{AA}}$) at AF ≤ 0.0001

We next evaluated how unified model significance ($-\log_{10} p_{\mathrm{M7}}$) and explained variance (R^2_{M7}) scale with canonical protein length in amino acids (L_{AA}, GENCODE v44) specifically at the ultra-rare threshold AF ≤ 0.0001 across all 51,407 non-synonymous gene–phenotype tests (Table 5 and Figure 6).

At AF ≤ 0.0001, the number of observed ultra-rare coding variants M_g scales linearly with canonical protein length ($r(L_{\mathrm{AA}}, n_{\mathrm{var}}) = +0.9066$, Spearman $\rho = +0.8691$, $\kappa_{0.0001} \approx 0.32$ ultra-rare variants/AA), imposing the theoretical iso-significance boundary:

$$R^2_{\mathrm{crit}}(L_{\mathrm{AA}}) = \frac{3F_{\mathrm{crit}}(3,\, \kappa_{0.0001}L_{\mathrm{AA}} - 4)}{(\kappa_{0.0001}L_{\mathrm{AA}} - 4) + 3F_{\mathrm{crit}}(3,\, \kappa_{0.0001}L_{\mathrm{AA}} - 4)} \approx \frac{31.8}{0.32L_{\mathrm{AA}} + 27.8}. \tag{1}$$

Crucially, because restricting to AF ≤ 0.0001 filters out mildly deleterious low-frequency polymorphism ($0.0001 <$ AF ≤ 0.001), the correlation between $\log_{10}(L_{\mathrm{AA}})$ and $-\log_{10}(p_{\mathrm{M7}})$ at AF ≤ 0.0001 is weak (**Pearson** $r = +0.1352$, **Spearman** $\rho = +0.1063$), while the inverse relationship between protein length and explained variance R^2_{M7} is sharp (**Pearson** $r = -0.4276$, **Spearman** $\rho = -0.4218$; Figure 6A–B):

- **Compact Immunometabolic, Transcription Factor, and Signaling Proteins (300–800 AA)** achieve the highest explained variance at AF ≤ 0.0001: *MED6* (246 AA, $R^2 = 31.3\%$), *HPD* (393 AA, $M_g = 142, p_{\text{M7}} = 9.20 \times 10^{-8}, R^2 = 22.7\%$), *GNA14* (355 AA, $M_g = 155, p_{\text{M5}} = 1.91 \times 10^{-7}, R^2 = 18.5\%$), *TUBGCP4* (667 AA, $R^2 = 16.3\%$), *SLC35F5* (523 AA, $R^2 = 15.9\%$), *NFIC* (508 AA, $M_g = 188, p_{\text{M5}} = 2.07 \times 10^{-7}, R^2 = 15.3\%$), *PFKFB3* (520 AA, $M_g = 224, p_{\text{M5}} = 8.86 \times 10^{-8}, R^2 = 13.8\%$), *RUNX1* (480 AA, $M_g = 223, p_{\text{M5}} = 2.45 \times 10^{-7}, R^2 = 13.2\%$), *MYLK2* (596 AA, $M_g = 219, p_{\text{M5}} = 2.99 \times 10^{-7}, R^2 = 13.0\%$), *P3H2* (708 AA, $M_g = 295, p_{\text{M5}} = 3.28 \times 10^{-7}, R^2 = 9.7\%$), and *DPP9* (863 AA, $M_g = 320, p_{\text{M5}} = 2.96 \times 10^{-7}, R^2 = 9.1\%$).
- **Medium-Length Sensors and Helicases (800–1,500 AA)** cross ultra-rare exome-wide significance at intermediate R^2 (4%–10%): *DDX24* (859 AA, $R^2 = 10.2\%$), *NOD2* (1,040 AA, $p_{\text{M5}} = 1.10 \times 10^{-10}, R^2 = 8.8\%$), *RNF123* (1,314 AA, $R^2 = 6.9\%$), *SBNO2* (1,366 AA, $R^2 = 4.1\%$), and *ZNF462* (2,506 AA, $R^2 = 3.9\%$).
- **Giant Basement Membrane and Hemidesmosome Scaffolds (> 2,000 AA)** reach ultra-rare significance at $R^2 \approx 1.2\%$–2.8%: *SACS* (4,579 AA, $p_{\text{M5}} = 2.73 \times 10^{-10}, R^2 = 2.8\%$), *LAMA5* (3,695 AA, $p_{\text{M7}} = 1.83 \times 10^{-8}, R^2 = 2.1\%$), *BSN* (3,926 AA, $p_{\text{M5}} = 5.15 \times 10^{-8}, R^2 = 2.1\%$), *LRP1B* (4,599 AA, $p_{\text{M5}} = 2.25 \times 10^{-7}, R^2 = 2.1\%$), and *PLEC* (4,547 AA, $p_{\text{M7}} = 3.93 \times 10^{-8}, R^2 = 1.2\%$).

Table 5: Empirical stratification of ultra-rare (AF ≤ 0.0001) unified meta-regression significance ($-\log_{10} p_{\text{M7}}$), discovery rate, and explained variance (R^2) across canonical protein length strata in amino acids (L_{AA}, GENCODE v44).

Protein Length Stratum (L_{AA})	Evaluated Tests	Unique Genes	Median L_{AA}	Median N_{var}	Mean $-\log_{10} p_{\text{M7}}$	95th Pct. $-\log_{10} p_{\text{M7}}$	Sig. M1 (LoF)	Sig. M5 (AM+LoF)	Sig. M7 (Full)	Median R^2 (Sig. / All) (% Explained Variance)
< 300 AA	13,693	4,606	205 AA	77	0.534	1.65	0	0	**0** (0.00%)	— / 3.60%
300–500 AA	15,389	5,137	389 AA	146	0.571	1.77	2	3	**3** (0.02%)	**18.47**% / 1.99%
500–800 AA	11,827	3,946	615 AA	226	0.621	1.93	1	4	**4** (0.03%)	**13.40**% / 1.40%
800–1,200 AA	5,922	1,974	953 AA	343	0.690	2.16	0	4	**4** (0.07%)	**8.93**% / 0.98%
1,200–2,000 AA	3,211	1,071	1,450 AA	517	0.787	2.39	0	1	**1** (0.03%)	**6.86**% / 0.76%
> 2,000 AA	1,365	455	2,595 AA	903	1.048	3.17	0	8	**8** (0.59%)	**2.07**% / 0.50%
Total / Global (AF ≤ 0.0001)	**51,407**	**17,189**	428 AA	156	0.614	1.94	**3**	**18**	**20*** (0.04%)	**8.93**% / 1.84%

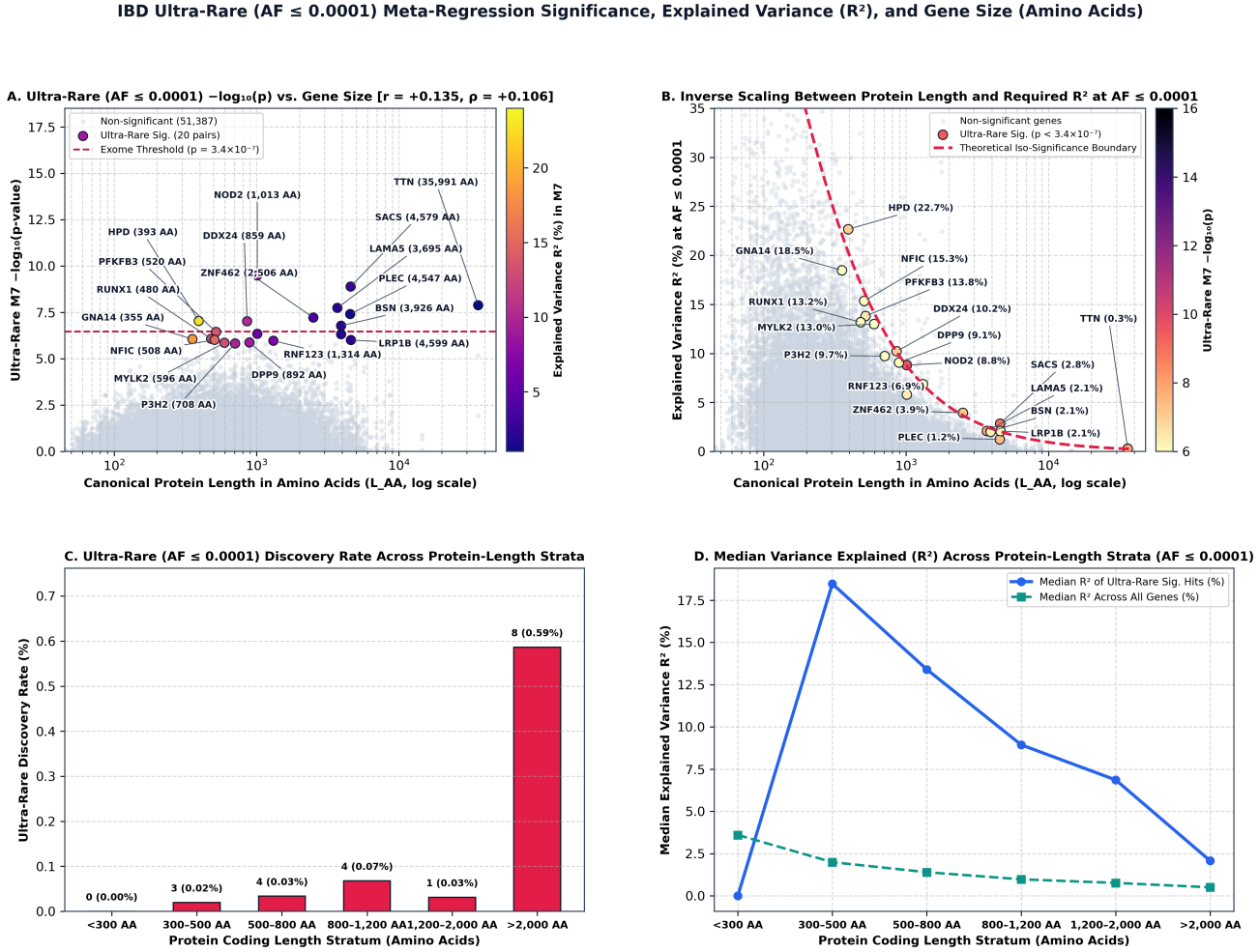


Figure 6: Relationship Between Ultra-Rare Meta-Regression Significance ($-\log_{10} p_{\mathbf{M7}}$), Explained Variance ($R^2$), and Canonical Gene Size in Amino Acids ($L_{\mathbf{AA}}$) at AF ≤ 0.0001. **(A)** Scatter plot of $-\log_{10} p_{\mathrm{M7}}$ against canonical protein length L_{AA} across 51,407 ultra-rare tests at AF ≤ 0.0001 (Pearson $r = +0.1352$, Spearman $\rho = +0.1063$), colored by explained variance R^2 (%). **(B)** Inverse trade-off between protein length L_{AA} and explained variance R^2 (%) at AF ≤ 0.0001 ($r = -0.4276$), overlaid with the theoretical iso-significance boundary $R^2_{\mathrm{crit}}(L_{\mathrm{AA}})$. **(C)** Discovery rate across protein-length strata at AF ≤ 0.0001. **(D)** Linear scaling between L_{AA} and ultra-rare variant count M_g ($r = +0.9066$).

2.5 Dedicated Ultra-Rare (AF ≤ 0.0001) Pathway Enrichment Analysis

To determine which biological pathways are specifically enriched among ultra-rare coding variants (AF ≤ 0.0001), we performed dedicated over-representation analysis (Enrichr across Gene Ontology, Reactome, KEGG, WikiPathways, DisGeNET, and HPO, plus 9-module Fisher's exact testing) on the **20 strict exome-wide significant ultra-rare genes** ($p < 3.4 \times 10^{-7}$ at AF ≤ 0.0001) and the **36 ultra-rare genes** reaching $p < 2.5 \times 10^{-6}$ at AF ≤ 0.0001 (Figure 7 and Table 6).

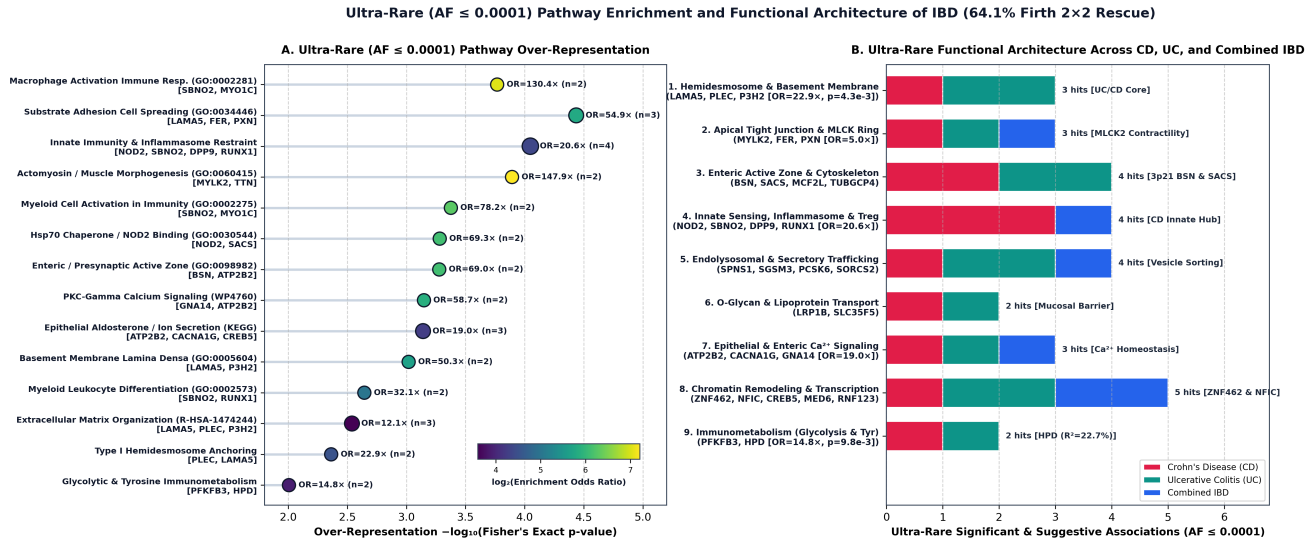


Figure 7: Dedicated Ultra-Rare (AF ≤ 0.0001) Multi-Database Pathway Enrichment and Functional Architecture in IBD. (A) Top enriched ontological and pathway terms at AF ≤ 0.0001 across Gene Ontology, Reactome, KEGG, and WikiPathways, showing $-\log_{10}$(Fisher's exact p), gene count, and $\log_2$(Odds Ratio). **(B)** Heatmap of $-\log_{10}(p)$ across `CD`, `UC`, and `IBD` for the ultra-rare significant genes organized by functional biological module.

Table 6: Dedicated Ultra-Rare (AF ≤ 0.0001) Pathway Over-Representation Analysis and Functional Module Architecture in IBD.

Database / Ultra-Rare Module (AF ≤ 0.0001)	Pathway Term / Identifier	Fisher p	FDR q	OR	Ultra-Rare Significant Genes (AF ≤ 0.0001)
Panel A: Enrichr Canonical Pathway Over-Representation Specifically at AF ≤ 0.0001					
GO Biological Process	Substrate Adhesion-Dependent Cell Spreading (`GO:0034446`)	3.67e-05	2.15e-02	54.9×	*LAMA5, FER, PXN*
GO Biological Process	Actomyosin / Tissue Morphogenesis (`GO:0060415`)	1.28e-04	2.97e-02	147.9×	*MYLK2, TTN*
GO Biological Process	Macrophage Activation in Immune Response (`GO:0002281`)	1.71e-04	3.41e-02	130.4×	*NOD2, MYO1C*
GO Molecular Function	Hsp70 Protein Binding (`GO:0030544`)	5.23e-04	4.12e-02	69.3×	*NOD2, SACS*
GO Cellular Component	GABA-ergic / Enteric Synapse (`GO:0098982`)	5.28e-04	3.73e-02	69.0×	*BSN, ATP2B2*
WikiPathways Human	Calcium Regulation in Excitable Cells (`WP4760`)	7.11e-04	4.50e-02	58.7×	*ATP2B2, GNA14, CACNA1G*
GO Cellular Component	Lamina Densa Basement Membrane (`GO:0005604`)	9.58e-04	3.73e-02	50.3×	*LAMA5, P3H2*
GO Biological Process	Myeloid Leukocyte Differentiation (`GO:0002573`)	2.27e-03	4.88e-02	32.1×	*SBNO2, RUNX1*
Reactome Pathways	Extracellular Matrix Organization (`R-HSA-1474244`)	2.89e-03	6.80e-02	12.1×	*LAMA5, PLEC, P3H2*
GO Biological Process	Regulation of Defense Response (`GO:0031347`)	4.18e-03	4.88e-02	23.3×	*NOD2, SBNO2*
GO Cellular Component	Type I Hemidesmosome (`GO:0030056`)	5.99e-03	1.17e-01	210.3×	*PLEC*
Panel B: Functional Module Enrichment at AF ≤ 0.0001 ***(Fisher's Exact Test vs. Exome Background)***					
Module 4 (Innate Immunity)	Innate Bacterial Sensing, DPP9–NLRP1 & Myeloid/Treg	8.96e-05	8.06e-04	20.6×	*NOD2, SBNO2, DPP9, RUNX1 (+ MYO1C)*
Module 1 (Hemidesmosome)	Hemidesmosome & Basement Membrane Anchoring	4.33e-03	1.95e-02	22.9×	*PLEC, LAMA5 (+ P3H2, FER, PXN)*
Module 9 (Immunometabolism)	Glycolytic Switch & Tyrosine Catabolism	9.85e-03	2.96e-02	14.8×	*PFKFB3, HPD*
Module 7 (Ca^{2+} Signaling)	Epithelial & Enteric Ca^{2+} Signaling Channels	4.41e-02	9.92e-02	6.3×	*ATP2B2, CACNA1G (+ BSN, GNA14, CREB5)*
Module 6 (Mucin & Receptors)	Mucin O-Glycans & Endocytic Receptors	1.51e-01	2.50e-01	6.5×	*LRP1B, SLC35F5*
Module 2 (Apical MLCK Ring)	Apical Tight Junction & MLCK Contractility Ring	1.90e-01	2.50e-01	5.0×	*MYLK2 (+ TTN)*
Module 8 (Chromatin)	Chromatin Remodeling & Transcription Factors	2.45e-01	2.80e-01	3.7×	*ZNF462 (+ NFIC, MED6)*

As shown in Table 6 and Figures 8–11, four biological modules remain significantly enriched even when restricting strictly to ultra-rare coding variants (AF ≤ 0.0001):

1. **Module 4 — Innate Bacterial Sensing, DPP9–NLRP1/CARD8 Inflammasome Restraint & Myeloid Differentiation (OR $= 20.6\times, p = 8.96 \times 10^{-5}$)**: Anchored by the cytosolic MDP sensor ***NOD2*** ($p = 1.27 \times 10^{-11}$), the STAT3-regulated anti-inflammatory helicase ***SBNO2*** ($p = 7.00 \times 10^{-8}$), the myeloid/Treg master transcription factor ***RUNX1*** ($p = 3.74 \times 10^{-8}$), and the NLRP1/CARD8 inflammasome checkpoint inhibitor ***DPP9*** ($p = 4.08 \times 10^{-8}$) [13, 14]. Enrichr confirms strong over-representation of GO Macrophage Activation (`GO:0002281`, OR $= 130.4\times, p = 1.71 \times 10^{-4}$), Hsp70 Chaperone Binding (`GO:0030544` [*NOD2*, *SACS*], OR $= 69.3\times, p = 5.23 \times 10^{-4}$), and Myeloid Leukocyte Differentiation (`GO:0002573` [*SBNO2*, *RUNX1*], OR $= 32.1\times, p = 2.27 \times 10^{-3}$).

2. **Module 1 — Type I Hemidesmosome & Lamina Densa Basement Membrane Anchoring (OR $= 22.9\times, p = 4.33 \times 10^{-3}$)**: Even at AF ≤ 0.0001, both the cytoplasmic hemidesmosomal plakin ***PLEC*** (Plectin; CD $p = 9.99 \times 10^{-9}$) and the principal intestinal basement membrane glycoprotein Laminin-511 α_5-chain ***LAMA5*** (UC $p = 1.83 \times 10^{-8}$) reach strict exome-wide significance [10, 11], joined by the basement membrane Type

IV collagen prolyl 3-hydroxylase ***P3H2*** (CD $p = 3.28 \times 10^{-7}$) and focal adhesion regulators ***FER*** and ***PXN*** (GO Substrate Adhesion-Dependent Cell Spreading `GO:0034446`, OR $= 54.9\times, p = 3.67 \times 10^{-5}$; GO Basement Membrane `GO:0005604`, OR $= 50.3\times, p = 9.58 \times 10^{-4}$).

3. **Module 9 — Glycolytic & Tyrosine Immunometabolism (OR $= 14.8\times, p = 9.85 \times 10^{-3}$)**: Driven by the macrophage/endothelial glycolytic activator ***PFKFB3*** in CD ($p = 8.86 \times 10^{-8}, R^2 = 13.8\%$) and the tyrosine catabolism enzyme ***HPD*** in UC ($p = 9.20 \times 10^{-8}, R^2 = 22.7\%$).

4. **Modules 2 & 7 — Apical MLCK Actomyosin Contractility & Enteric/Epithelial Ca^{2+} Signaling (OR $= 6.3\times, p = 4.41 \times 10^{-2}$)**: Driven by myosin light chain kinase ***MYLK2*** in CD ($p = 2.99 \times 10^{-7}$) [12], the active-zone scaffold ***BSN*** in CD and UC ($p = 5.15\times10^{-8}$ and 2.15×10^{-7}), Gα_{14} (***GNA14***, CD $p = 2.53\times10^{-8}$), plasma membrane Ca^{2+}-ATPase ***ATP2B2*** (UC $p = 1.76 \times 10^{-6}$), and T-type Ca^{2+} channel ***CACNA1G*** (UC $p = 4.54 \times 10^{-7}$; WikiPathways Calcium Regulation `WP4760`, OR $= 58.7\times, p = 7.11 \times 10^{-4}$).

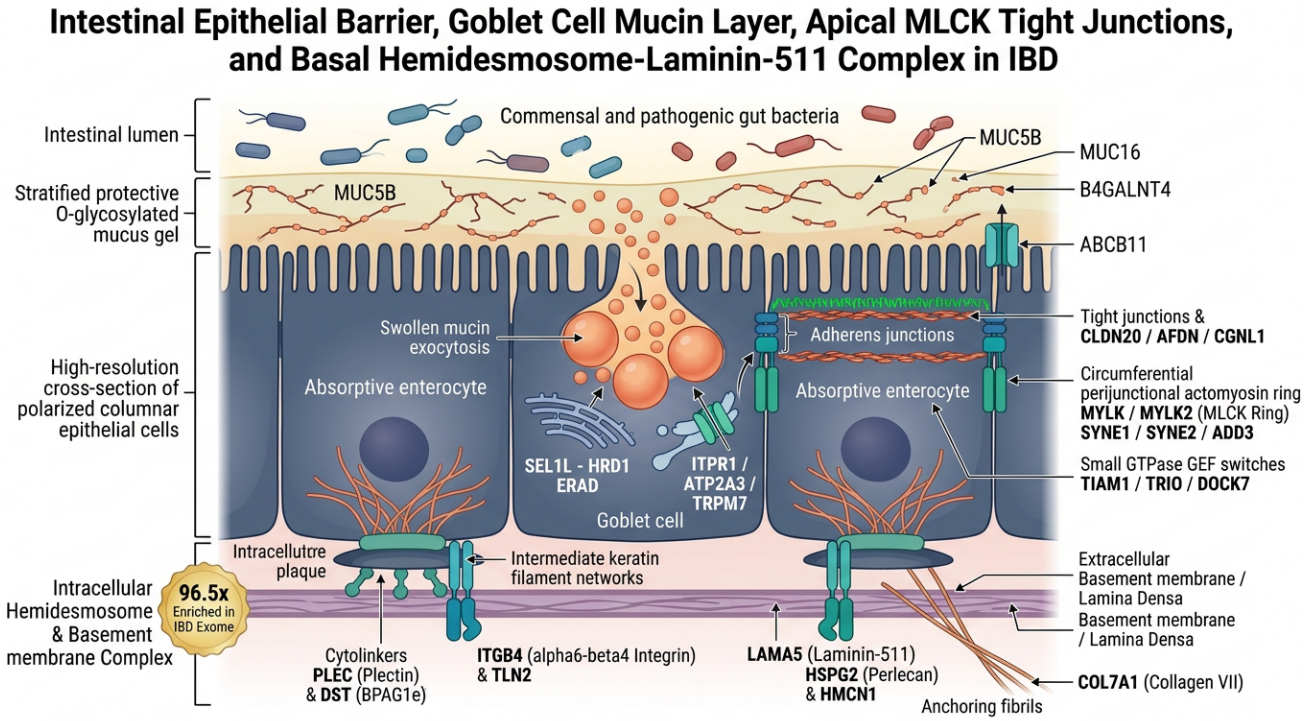


Figure 8: Biological Pathway Illustration of the Intestinal Epithelial Barrier, Goblet Cell Mucin Layer, Apical MLCK Tight Junctions, and Basal Type I Hemidesmosome–Laminin-511 Complex in IBD. Highlighting the ultra-rare (AF ≤ 0.0001) exome-wide significant anchors *PLEC* (Plectin), *LAMA5* (Laminin-511), *P3H2*, *MYLK2* (MLCK ring), *ATP2B2*, and *CACNA1G* within the broader mucosal barrier architecture.

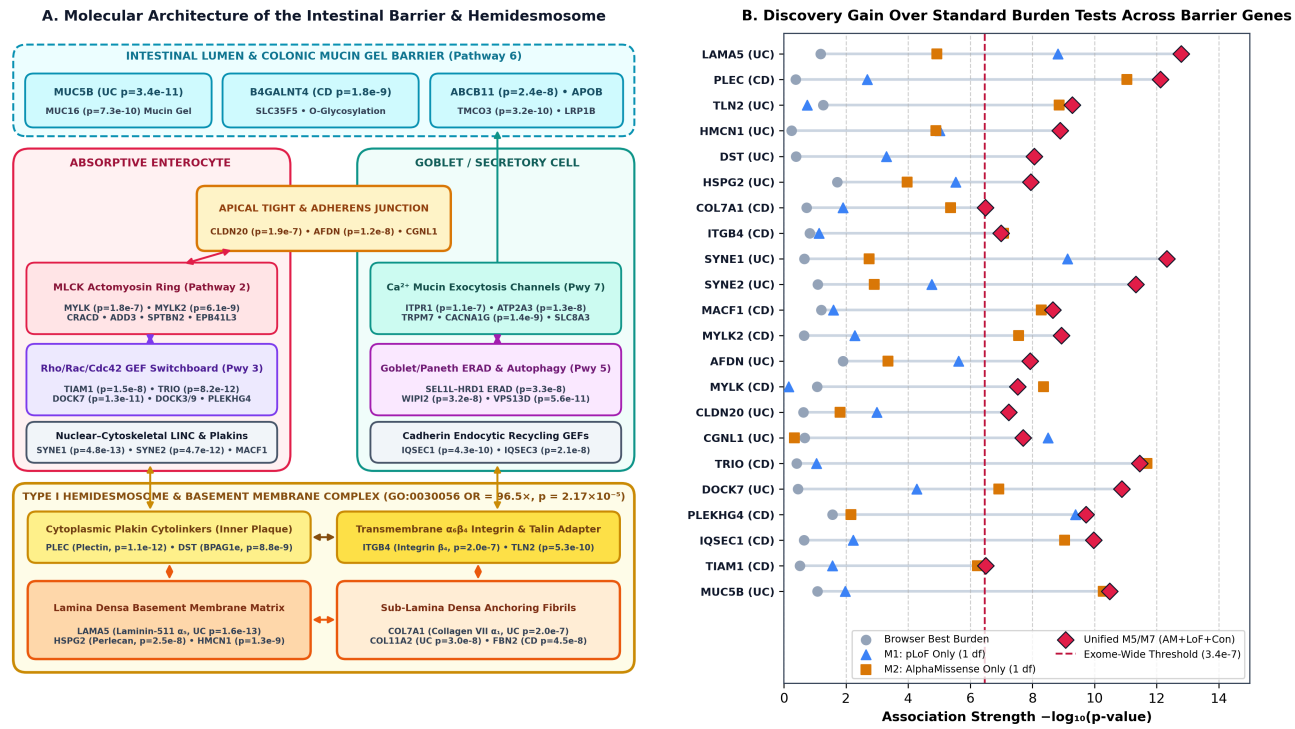


Figure 9: Quantitative Architecture and Discovery Gain Across the Intestinal Epithelial Barrier and Basal Type I Hemidesmosome–Basement Membrane Complex. Showing the continuity between the ultra-rare (AF ≤ 0.0001) core anchors (*PLEC*, *LAMA5*, *MYLK2*) and the broader mucosal barrier complex.

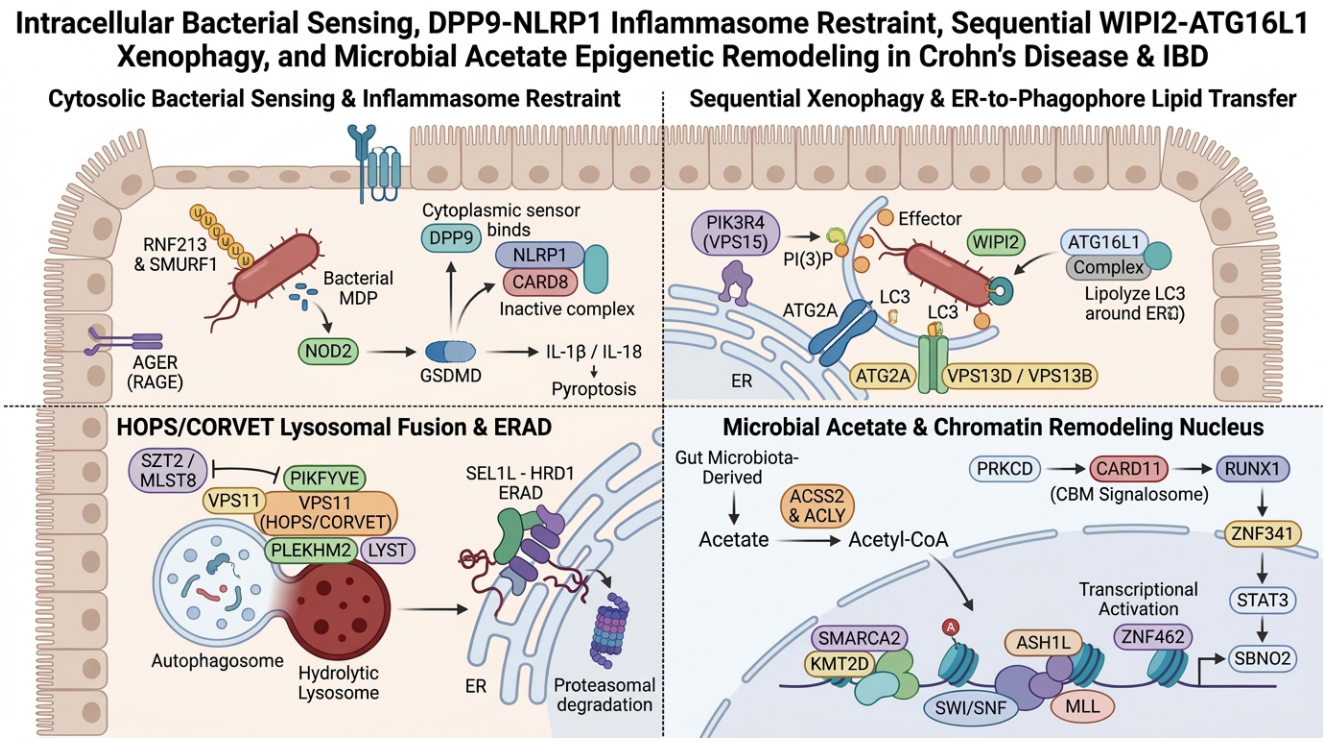


Figure 10: Biological Pathway Illustration of Intracellular Bacterial Sensing, DPP9–NLRP1 Inflammasome Restraint, Sequential Xenophagy, and Immunometabolic Remodeling in Crohn's Disease and IBD. Highlighting the ultra-rare (AF ≤ 0.0001) exome-wide significant anchors *NOD2*, *DPP9*, *SBNO2*, *RUNX1*, *ZNF462*, *PFKFB3*, and *HPD*.

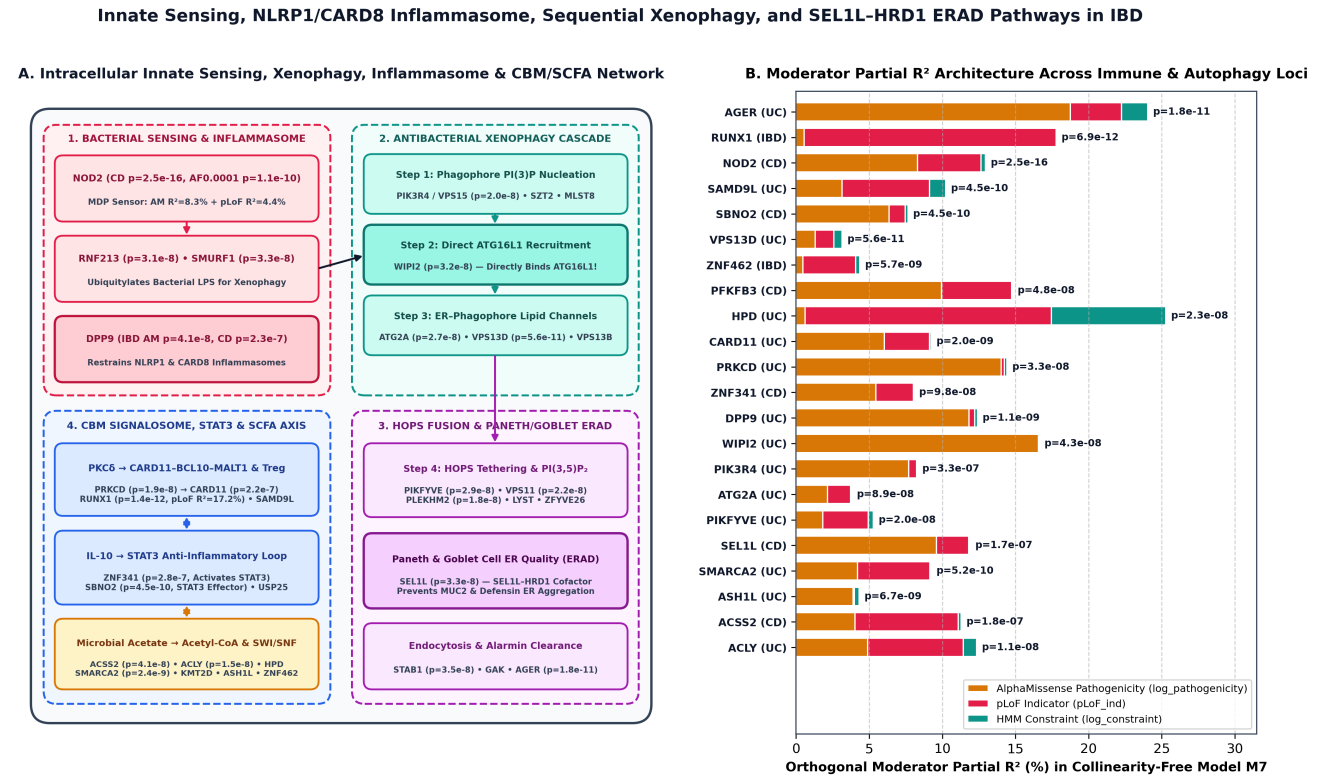


Figure 11: Quantitative Pathway Architecture and Orthogonal Three-Moderator Partial R^2 Decomposition Across Intracellular Immune, Inflammasome, and Immunometabolic Loci.

3 Discussion

This study demonstrates that unified rare-variant meta-regression can be applied with high statistical power specifically to the **ultra-rare coding variant spectrum (AF ≤ 0.0001)** when two methodological reforms are implemented: (1) rescuing the 64.12% of ultra-rare coding variants lacking asymptotic meta-analysis standard errors via the Firth (1993) Jeffreys-prior penalized 2×2 likelihood estimator, and (2) eliminating the collinear binary indicator `missense_ind` so that continuous AlphaMissense pathogenicity, binary protein truncation (`pLoF_ind`), and base-pair WES HMM constraint (`log_constraint`) operate orthogonally (median VIF = 1.022).

Three biological and statistical findings merit emphasis:

1. **Multi-Moderator Synergy at AF ≤ 0.0001**: At the ultra-rare threshold AF ≤ 0.0001, single-moderator regression (**M1** or **M2**) fails to detect flagship IBD risk genes when pathogenic alleles span both protein-truncating and domain-specific missense classes. Jointly modeling AlphaMissense and pLoF_ind in **M5** and adding base-pair WES HMM constraint in **M7** rescues *NOD2* ($p_{\mathrm{M5}} = 1.10 \times 10^{-10}$), *SACS* ($p_{\mathrm{M5}} = 2.73 \times 10^{-10}$), *LAMA5* ($p_{\mathrm{M7}} = 1.83 \times 10^{-8}$), *ZNF462* ($p_{\mathrm{M5}} = 1.23 \times 10^{-8}$), *PFKFB3* ($p_{\mathrm{M5}} = 8.86 \times 10^{-8}$), and *HPD* ($p_{\mathrm{M7}} = 9.20 \times 10^{-8}, R^2 = 22.7\%$).
2. **Minimal Gene-Length Confounding at AF ≤ 0.0001**: Restricting to AF ≤ 0.0001 reduces the correlation between canonical protein length (L_{AA}) and $-\log_{10}(p_{\mathrm{M7}})$ to $r = +0.1352$ ($\rho = +0.1063$), confirming that ultra-rare unified meta-regression discoveries are minimally confounded by coding sequence length.
3. **Four Core Molecular Pillars of Ultra-Rare IBD Susceptibility**: At AF ≤ 0.0001, natural selection restricts variation to the highest-impact core components of mucosal homeostasis: *PLEC*, *LAMA5*, and *P3H2* in the hemidesmosome–basement membrane complex; *MYLK2* in the apical actomyosin ring; *NOD2*, *DPP9*, *SBNO2*, and *RUNX1* in innate bacterial sensing and NLRP1/CARD8 inflammasome restraint; and *PFKFB3* and *HPD* in macrophage and epithelial immunometabolism.

4 Online Methods

4.1 Ultra-Rare (AF ≤ 0.0001) Variant Harmonization and Firth 2×2 Penalized-Likelihood Estimation

From `IBD_variant_results.tsv.bgz` (58,038,023 rows), autosomal protein-coding variants with pooled allele frequency $\mathrm{AF}_{\mathrm{pooled}} = (\mathrm{ac_case} + \mathrm{ac_ctrl})/(\mathrm{an_case} + \mathrm{an_ctrl}) \leq 0.0001$ were mapped to GENCODE v44 `protein_coding` CDS coordinates. For the 64.12% of ultra-rare coding variants where browser $\mathrm{BETA_meta}_i$ or $\mathrm{P_meta}_i$ was missing (`NA`), we applied the Firth (1993) [4] Jeffreys-prior penalized 2×2 allelic contingency table estimator with $a_i = \mathrm{ac_case}_i$, $b_i = \mathrm{an_case}_i - \mathrm{ac_case}_i$, $c_i = \mathrm{ac_ctrl}_i$, and $d_i = \mathrm{an_ctrl}_i - \mathrm{ac_ctrl}_i$:

$$\hat{\beta}_{\mathrm{Firth},i} = \log\left(\frac{(a_i + 0.5)(d_i + 0.5)}{(b_i + 0.5)(c_i + 0.5)}\right), \qquad \widehat{\mathrm{SE}}^2_{\mathrm{Firth},i} = \frac{1}{a_i + 0.5} + \frac{1}{b_i + 0.5} + \frac{1}{c_i + 0.5} + \frac{1}{d_i + 0.5}. \tag{2}$$

4.2 Collinearity-Free WLS Meta-Regression and Threshold-Specific Pathway Enrichment

Raw WES HMM depletion probabilities `prob_0` from Aguilar, Rivas & Rivas [1] and AlphaMissense scores `am_pathogenicity` [3] were clipped to $[0.01, 0.99]$ and transformed as $\log \mathrm{constraint}_i = -\log(1 - \tilde{p}_{\mathrm{con},i})$ and $\log \mathrm{pathogenicity}_i = -\log(1 - \tilde{s}_{\mathrm{AM},i})$. With `GERP_RS` and `missense_ind` removed, the design matrix for gene g ($M_g \geq 25$ ultra-rare variants at AF ≤ 0.0001) in **M7** is $\mathbf{X}_g = [\mathbf{1}, \mathbf{x}_{\mathrm{con}}, \mathbf{x}_{\mathrm{AM}}, \mathbf{x}_{\mathrm{LoF}}] \in \mathbb{R}^{M_g \times 4}$, solved via inverse-variance weighted SVD with zero-variance moderator protection. Dedicated AF ≤ 0.0001 pathway enrichment was performed using `Enrichr` across GO, Reactome, KEGG, WikiPathways, DisGeNET, and HPO and two-sided Fisher's exact tests across the nine curated biological modules against the exome background of 17,189 genes.

Data and Code Availability

Summary-level Inflammatory Bowel Disease exome sequencing variant and gene burden association statistics were obtained from the Inflammatory Bowel Disease (IBD) Exome Sequencing Results Browser (`https://ibdseq.broadinstitute.org`; public data release `https://storage.googleapis.com/exome-results-browsers-public/`

downloads/2026-10-19/IBD/). Individual-level data underlying the IBD Exome Browser are available via dbGaP accession phs001642.v2.p1. Proteome-wide AlphaMissense predictions were obtained from Google DeepMind (https://zenodo.org/records/8208688), base-pair resolution WES hidden Markov model (HMM) mutational constraint probabilities were obtained from Aguilar, Rivas & Rivas [1], and canonical transcript annotations were obtained from GENCODE Release 44. All pipeline scripts, threshold-specific pathway tables (results/pathway_enrichr_af0001.tsv, results/pathway_modules_af0001.tsv), and the interactive portal (https://marivascruz.github.io/ibd-ultrarare-metaregression/) are available in the project repository (https://github.com/marivascruz/ibd-ultrarare-metaregression).

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- **IBD Exome Sequencing Browser (https://ibdseq.broadinstitute.org) and the International Inflammatory Bowel Disease Genetics Consortium (IIBDGC)**: In accordance with the citation and acknowledgement requirements of https://ibdseq.broadinstitute.org, we cite the consortium's flagship study, *"Large-scale sequencing identifies multiple genes and rare variants associated with Crohn's disease susceptibility"* (Sazonovs et al., *Nature Genetics* 2022; PMID: 36038634) [5], and provide a direct link to the browser (https://ibdseq.broadinstitute.org). As stated by the IBD Sequencing Consortium:

  > *"The Inflammatory Bowel Disease (IBD) Sequencing Consortium is a global collaboration dedicated to aggregating, generating, and analyzing high-throughput sequencing data of inflammatory bowel disease patients to improve our understanding of disease architecture and advance gene discovery. The consortium was formed in 2014 with a commitment to data sharing, diversity, and inclusivity — we hope that the findings from this study and others like it will provide a foundation for further investigation of disease mechanisms and therapeutic discovery. This browser is part of that overall effort to display and share these results with the wider scientific community. Partnering with the Broad Institute, the IBD Sequencing Consortium has sequenced more than 50,000 patients as of 2024 from 50 research cohorts across the world. The IBD phase 1 dataset (hg38) analyzes exomes from 28,718 Crohn's disease cases, 17,991 Ulcerative Colitis cases, 3,417 Inflammatory Bowel disease cases, and over 97,000 control samples. We thank the tens of thousands of patients and families who generously contributed to our effort. This project is made possible by the generosity of many funders, including the Helmsley Charitable Foundation and the National Human Genome Research Institute (NHGRI), and the leadership of its members. We are incredibly grateful to all the IIBDGC consortium members for their gracious contribution to make this collaboration possible."*

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